

Pooja Kiran Bharadwaj

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EDUCATION

Arizona State University

Tempe, AZ

Master of Science in Information Technology

Aug. 2024 – May 2026

- Coursework: Machine Learning Security, Cloud Security Architecture, Information Assurance, Network Defense, Secure Systems

M. S. Ramaiah University of Applied Sciences

Bengaluru, India

Bachelor of Technology in Computer Science and Engineering

Aug. 2019 – Aug. 2023

EXPERIENCE

AI Security Researcher & Engineer

Aug. 2024 – Present

Independent Security Research (Concurrent with M.S. program)

Tempe, AZ

- Engineered an inline MCP security gateway with a 5-layer decision pipeline and 50+ prompt-injection detection rules by implementing JSON-RPC parsing, Unicode/homoglyph normalization, and real-time policy enforcement in Python/FastAPI, achieving 0.015ms average inspection latency per tool call across 50,000 benchmarked iterations
- Developed a static IAM policy analyzer for AI agent roles on AWS by implementing 25 deterministic rules covering wildcard permissions, privilege escalation, audit-trail tampering, and credential-harvest chains with SARIF output, enabling automated least-privilege enforcement as a CI merge gate with zero runtime dependencies
- Built a model supply-chain scanner with a taint engine and symbolic resolver covering multiple model file formats by implementing deep opcode analysis for pickle Protocol 0–5, SafeTensors header inspection, GGUF structural parsing, and typosquat detection, achieving 100% detection on 12 attack fixtures (verified CVEs) in 116ms with 0% false positive rate
- Created adversarial prompt-generation and model-extraction attack harnesses (FGSM, PGD, C&W) mapped to MITRE ATLAS by implementing multi-category red-team generation and detector evaluation with structured SARIF evidence output, producing reproducible security assessments for guardrail validation and input/output filtering decisions

Business Compliance Lead & Market/Cost Analyst

Aug. 2025 – Dec. 2025

AEROSEC – ASU Technology Innovation Lab (Honeywell Aerospace)

Tempe, AZ

- Led regulatory compliance analysis for Honeywell Aerospace's security product by mapping requirements and conducting competitive cost modeling across aerospace frameworks, delivering a compliance framework supporting the go-to-market strategy

Graduate Teaching Assistant / IT Grader

Jan. 2025 – Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Tempe, AZ

- Evaluated technical assignments for 100+ students per semester by assessing cloud security architectures, network defense configurations, and IAM access control implementations, maintaining consistent grading standards across multiple course sections

PROJECTS

MCP Agent Security Gateway | *Python, FastAPI, Docker, Kubernetes, GitHub Actions*

- Architected an inline stdio proxy intercepting MCP tools/call JSON-RPC requests through a 5-layer pipeline (server trust, tool-call policy, process-spawn evaluation, semantic intent, network egress) by implementing allow/block/redact/quarantine decisions, preventing unauthorized tool execution before requests reach downstream servers
- Implemented 50+ prompt-injection detection rules with Unicode and homoglyph normalization by building PII detection for 9 data types and exfiltration-pattern matching, achieving comprehensive coverage across instruction override, role manipulation, delimiter injection, and encoded payload attacks
- Engineered production-grade operational controls by integrating SHA-256 hash-chained audit logging, circuit breakers, rate limiting, shadow mode, and write-ahead logging with Kubernetes deployment manifests, providing tamper-evident observability with full traceability across all security decisions

- Validated security controls with 529 automated tests at 77% coverage across Python 3.10/3.11/3.12 by integrating Bandit, CodeQL, Trivy, Gype, and pip-audit into the CI pipeline, achieving p50 latency of 0.0123ms and p99 of 0.04ms with near-zero performance overhead

AWS Agent Identity Guard | *Python, AWS IAM, Terraform, SARIF*

- Developed 25 deterministic IAM rules targeting AI agent roles by detecting wildcard service prefixes, iam:PassRole without conditions, privilege-management actions, audit-trail tampering, and credential-harvest plus lateral-movement chains, covering the full agent identity attack surface for Bedrock, SageMaker, and Lambda workloads
- Implemented credential-harvest chain detection by analyzing combined permission patterns across sts:AssumeRole, cross-account trust without ExternalId, and network egress modifications, identifying multi-step privilege escalation paths in a single policy scan
- Built a CI merge gate by outputting findings in Text, JSON, and SARIF formats with exit code 1 on critical/high severity, integrating with GitHub Code Scanning for inline PR annotations that block deployment of over-permissioned agent roles
- Achieved zero runtime dependencies for local policy scanning by implementing pure-Python static analysis with optional boto3 live-account mode, Terraform infrastructure examples, and documented threat model with security audit

HF Model Provenance Scanner | *Python, Hugging Face API, Docker, SARIF*

- Engineered a taint engine and symbolic resolver for pickle opcode analysis (Protocol 0–5) by implementing deep REDUCE/GLOBAL/STACK_GLOBAL walks with gadget-chain matching, detecting memoized-global exec and importlib bypass techniques that industry competitor ModelScan 0.8.8 missed
- Built temporal baseline diffing for rug-pull detection by implementing SafeTensors header inspection, GGUF structural parsing, typosquat detection via Levenshtein distance, and multi-layer obfuscation decode (base64, chr(), string concat), covering 9 distinct attack vectors across the model supply chain
- Reduced model scan bandwidth by 99.9% by using HTTP Range requests to fetch only first 512KB of opcodes and metadata headers up to 16MB, scanning the full GPT-2 repository in 0.5MB instead of downloading 500MB of tensor weights
- Detected 12/12 attack fixtures (CVE-2024-5480, CVE-2024-25664) in 116ms total with 0/5 false positives by supporting 8 file format families, CycloneDX 1.5 SBOM generation, MITRE ATT&CK v19 mapping, and CI templates across 5 platforms

LLM Red-Team Framework | *Python*

- Built an adversarial prompt-generation and detector-evaluation framework by implementing multi-category attack generation following OWASP LLM Top 10, producing per-category detection-rate reports with structured JSON evidence for guardrail validation

PUBLICATION

A Personalized E-Learning System Using Reinforcement Learning Through Satellite

IEEE INDICON 2023 | IEEE Xplore: 10440852

TECHNICAL SKILLS

AI/ML Security: Agentic AI Security, LLM/RAG Security, Prompt Injection & Jailbreak Testing, MCP/Tool Security, Model Supply-Chain Security, Adversarial ML (FGSM, PGD, C&W), Data Poisoning, Membership Inference, Model Extraction, Differential Privacy

Standards: MITRE ATT&CK/ATLAS, OWASP Top 10 for LLM Applications, NIST AI RMF, CycloneDX SBOM

Cloud & IAM: AWS IAM, Least-Privilege Enforcement, Trust Relationships, Agent Role Security, Cloud Security, Terraform

Languages & Tools: Python, FastAPI, PyTorch, scikit-learn, Docker, Kubernetes, GitHub Actions, CI/CD, SARIF

Security Eng.: Threat Modeling, Red Teaming, Guardrails Design, Input/Output Filtering, Access Controls, Audit Logging, Circuit Breakers

CERTIFICATIONS & RECOGNITION

AWS Academy Graduate: Cloud Security Foundations | **Technology Innovation Lab:** Honeywell Aerospace & ASU | **KSCST:** Research Grant